Introduction to Focus Areas – Data Science – WS24/25

Lecturer: Katharina Jahn

Assignment 1

- Deadline Oct 28, 10:00
- Upload your notebook (Jupyter or R notebook) and a pdf created from the notebook to the whiteboard
- State the group number and the names of all group members

The Data

We will use the Heart Disease Data Set „The Cleveland Database“, see https://archive.ics.uci.edu/ml/datasets/Heart+Disease)

You find a copy of the dataset on the Whiteboard.

This database contains clinical and bio-medical attributes for more than 300 patients. The "goal" field refers to the presence of a heart disease in a patient. It is integer valued from 0 (no presence) to 4.

The Tasks

1. Perform an exploratory data analysis on the Cleveland heart disease dataset described above. You can reuse your code from the in-class project. Add at least two new plot types like pairplot and/or heatmap from the lecture (e.g. from the seaborn library in python).
2. Train at least three types of classifiers to diagnose heart disease based on the available data and compare them visually in a grouped bar chart with regard to accuracy, sensitivity and specificity. Create a confusion matrix for each classifier and plot them as a heatmap.
3. Repeat the steps of the previous task using as class labels only “heart disease vs. no heart disease”. Why does this change make sense?
4. Create and plot ROC curves for your classifiers (2-class problem only).